# BITWISE OPERATORS

Bitwise operator's works on bits.

Turbo-c is a 16 bit compiler. Due to this bitwise operations are limited to 16 bits only [$2^0$ to $2^{15}$].

Bitwise operators operate **integer** type values only.

We have to calculate only the **on** bits [ **1** ].

When the first bit[**Sign bit**] is **1** then the number is **Negative** and it is **0** then the number is **positive**.

They are very much used in system software development.

**Note: Bitwise operator is low level feature.**

C-Language supports following bitwise operators.

**&** -Bitwise and

**|** - Bitwise or

**^** - XOR ==> Exclusive OR

**~** - Compliment operator

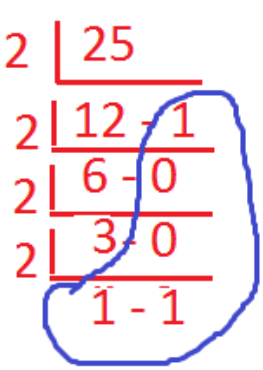
**<<** - Left shift operator

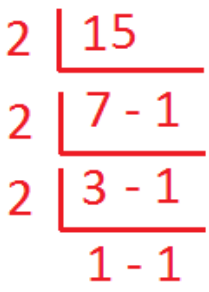
**>>** - Right shift operator

**& - Bitwise and:** In this both bits are 1's then result bit is 1. Otherwise result bit is 0.

Eg: **25 & 15 = 9**

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111




25 & 15 = 9

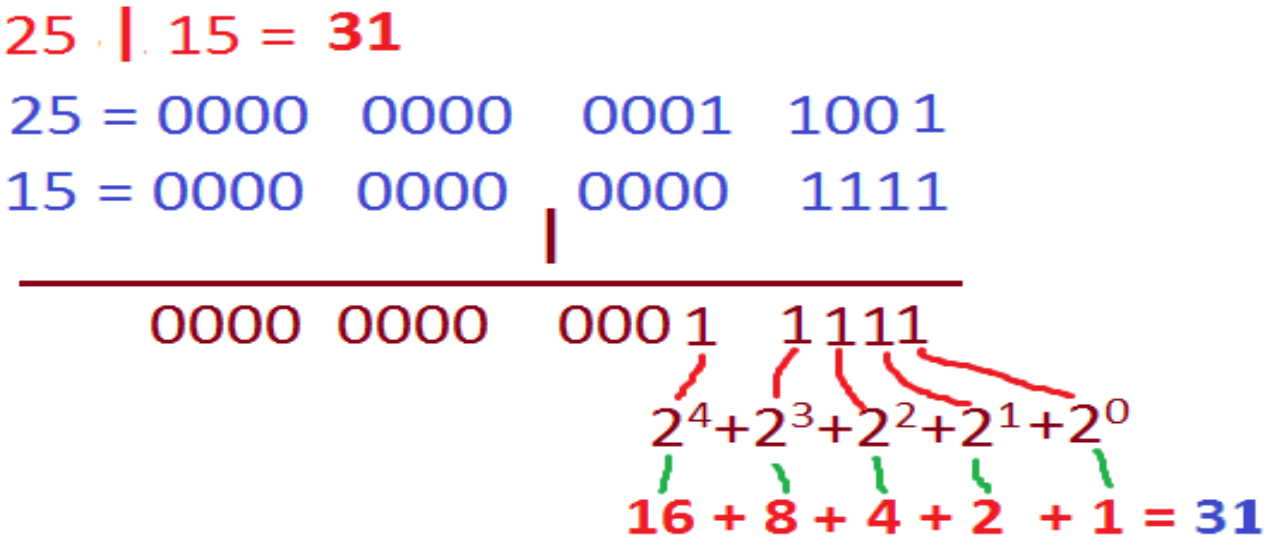
25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111
&
0000 0000 0000 1001

$2^3 + 2^0$

8 + 1 = **9**

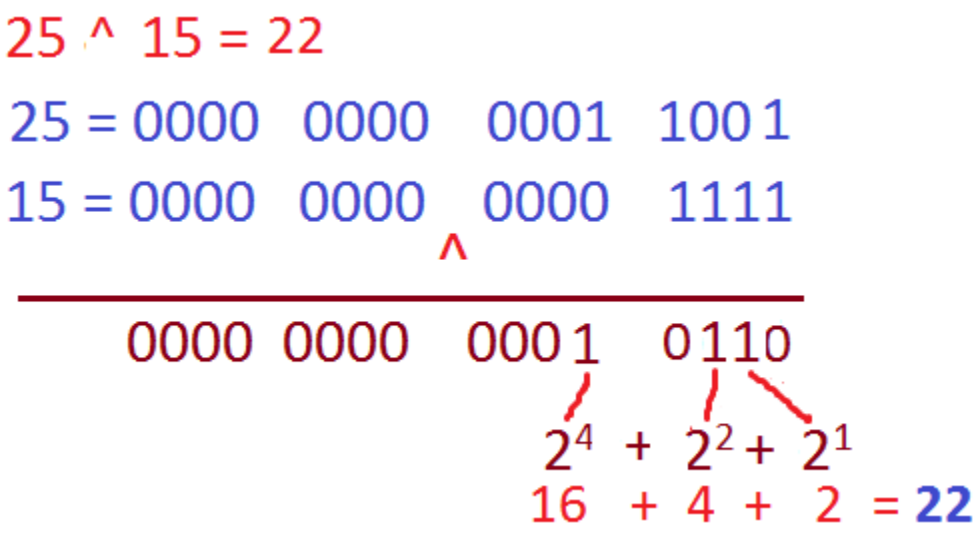
**| - Bitwise or**: In this both bits are 0's then result bit is 0. Otherwise result bit is 1.

Eg: 25 | 15 = 31



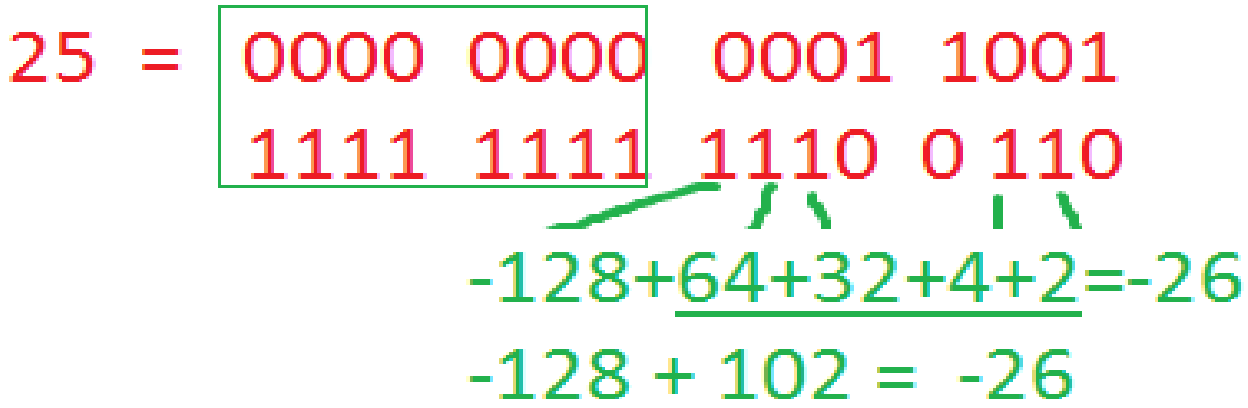
**^ - XOR [Exclusive OR]**: In this both bits are same then result bit is 0. Otherwise result bit is 1.

Eg: 25 ^ 15 = 22



**~ - Compliment operator:** In compliment operation the bits are complimented. i.e. 1's become 0's and 0's become 1's. Due to this +Ve no becomes –Ve and –Ve no becomes +Ve.

eg: **~25 ➔ -26**



25 = 0000 0000 0001 1001

~ = 1111 1111 1110 0110

2+4+32+64+128+256+512+1024+2048+4096+8192+16384-32768=-26

~-25 = 0000 0000 0001 1001

1's ~ = 1111 1111 1110 0110

2's ~ = 0000 0000 0000 0001

1111 1111 1110 0111

16+8=24

1 0 0 01

0 1 0 1

1 1 0 10

**Note: When starting bit is 1 given no is –Ve.**

**Eg: ~-25 ➔ +24**

~-25 = +24

25 = 0000 0000 0001 1001

1111 1111 1110 0110 <== 1's compliment

+1 <== 2's Compliment

1111 1111 1110 0111

~ 0000 0000 0001 1000

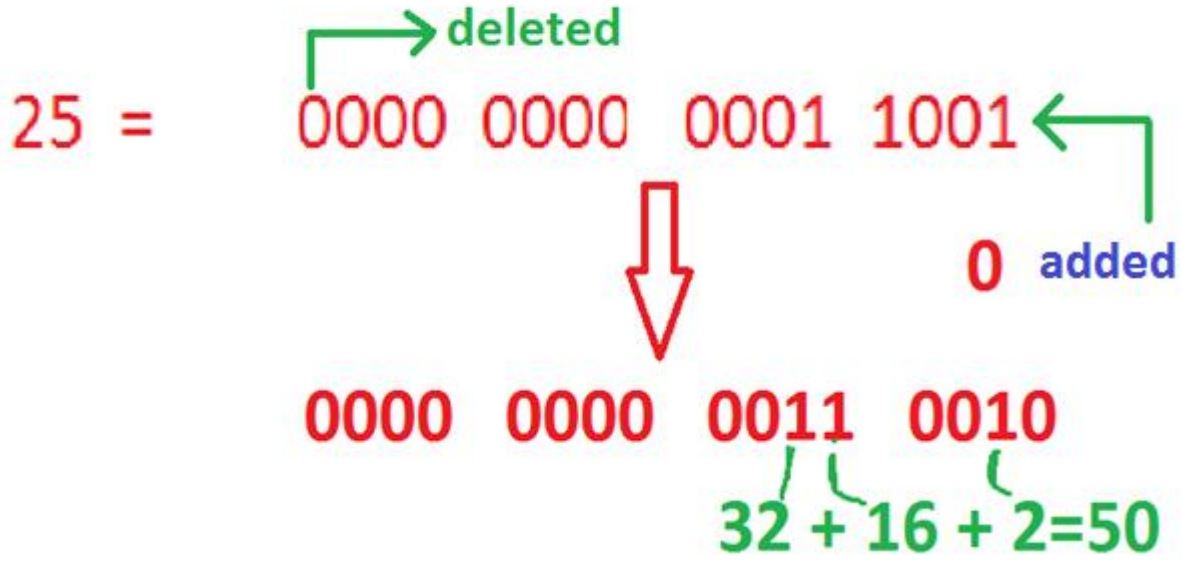
16+8=24

## << - left shift operator:

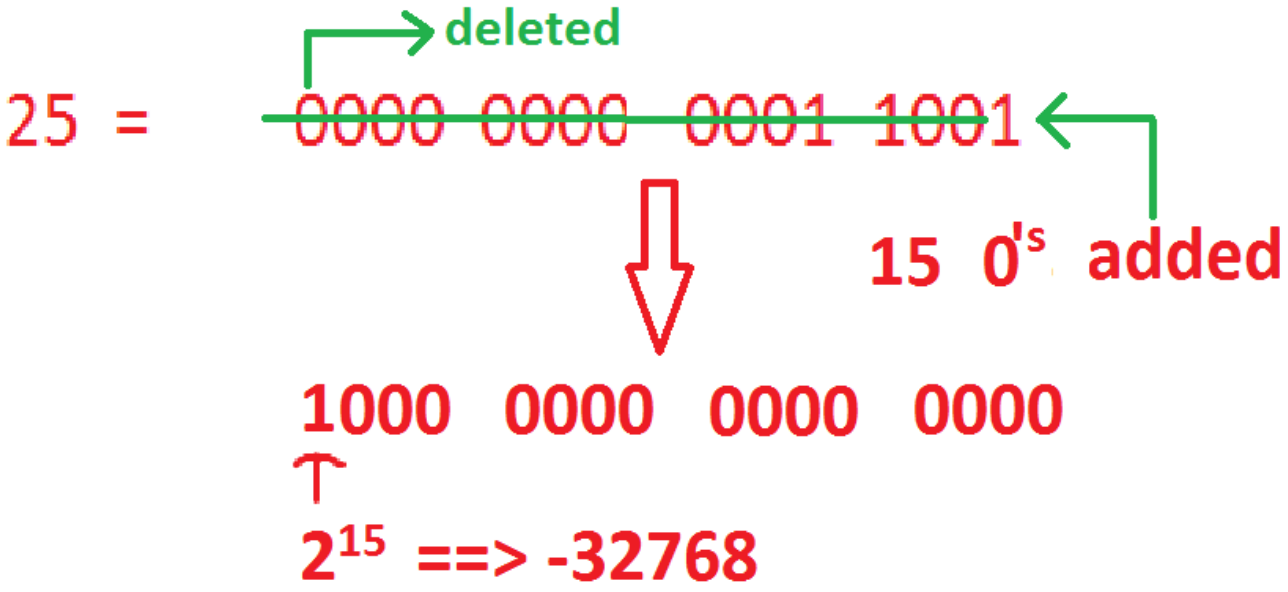
**In left shift operation, the specified no of bits are deleted from left side and the same no of zeros added on right side. In left shift operation, most probably the value is multiplied with 2 that no of times.**

**Eg:25<<1=50, 25<<2=100, 25<<15 =-32768, 25<<16=0**

**eg: 25<<1=50**


deleted
25 =
0000 0000 0001 1001
0 added
0000 0000 0011 0010
32 + 16 + 2=50

eg: 25<<15 = -32768



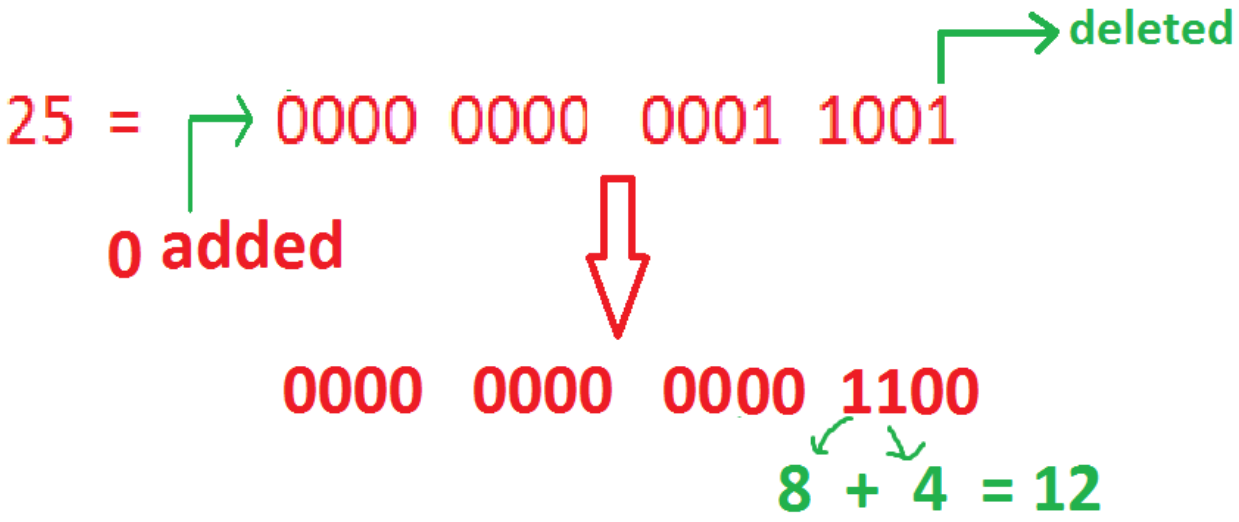

Note: When starting bit 1 no is negative.

## >> - Right shift operator:

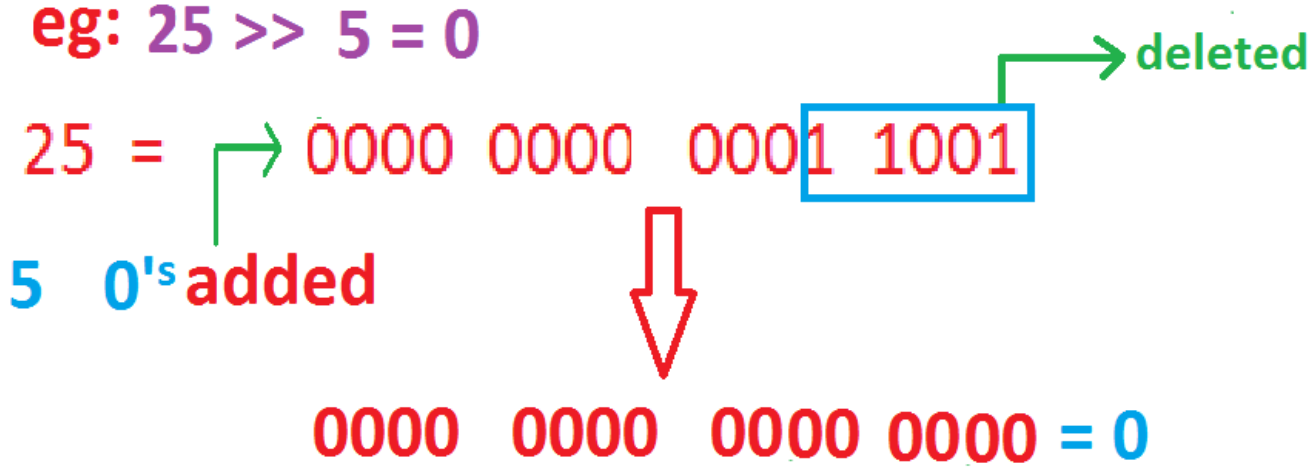
**In right shift operation, the bits are moved to right side i.e. the specified no.of bits are deleted from right side and same no.of zero's are added left side. Due to this always the number is divided with 2 that no of times.**

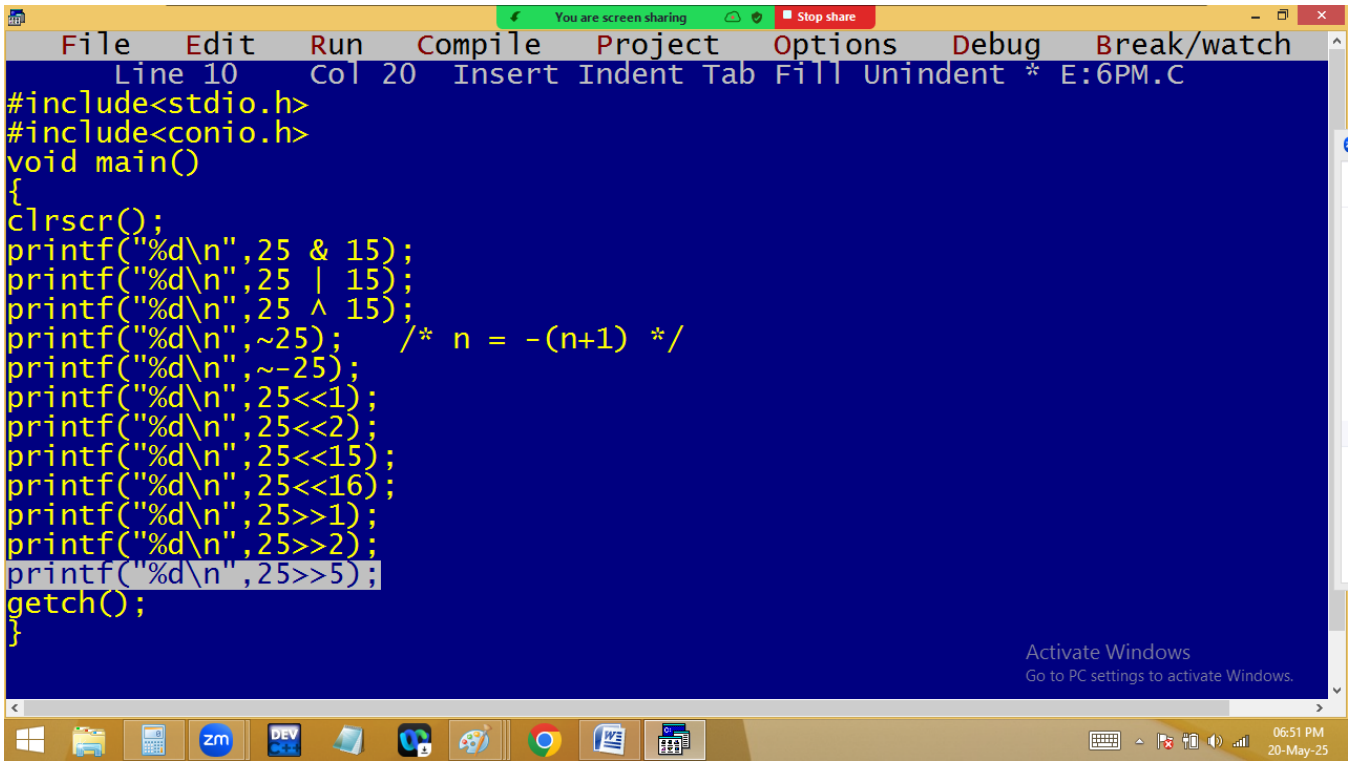
**Eg:25>>1=12, 25>>2=6, 25>>3=3, 25>>4=1,25>>5=0**

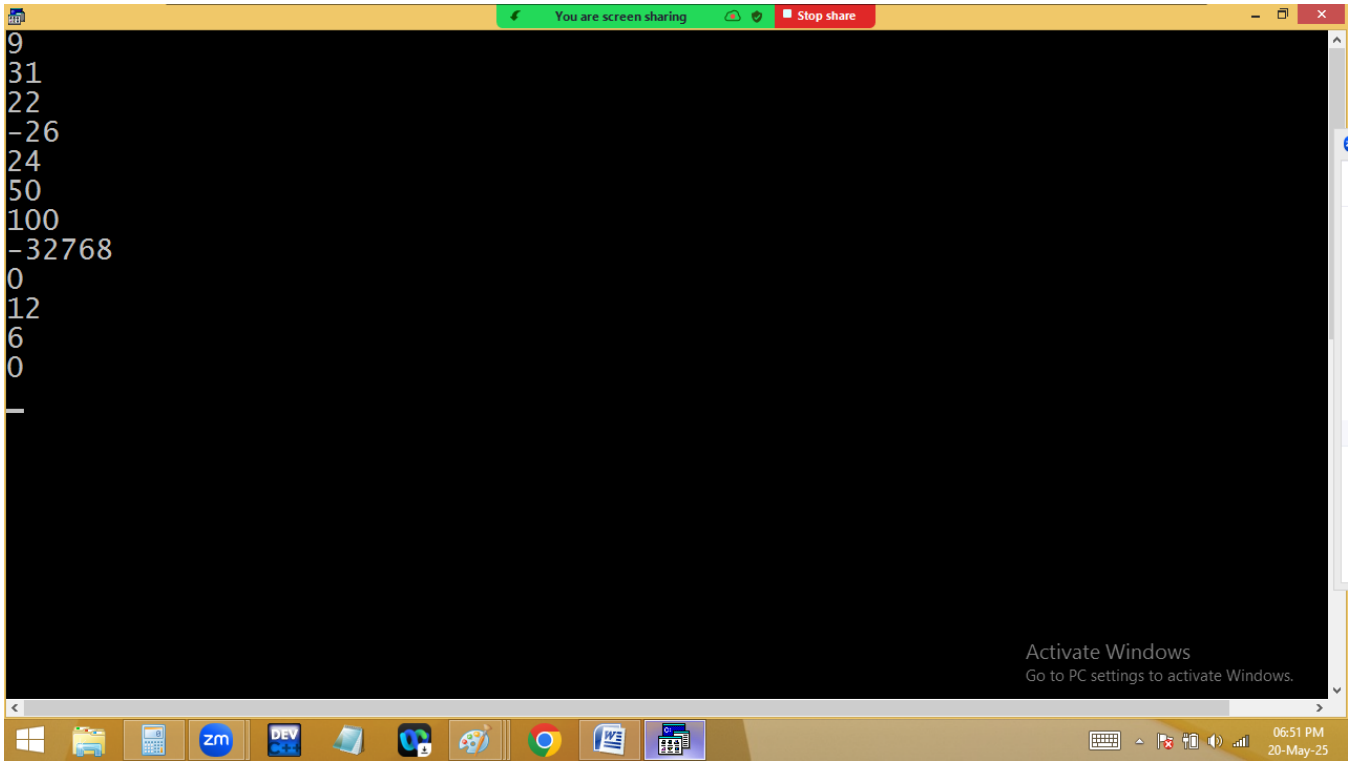
eg: 25 >>1 = 12

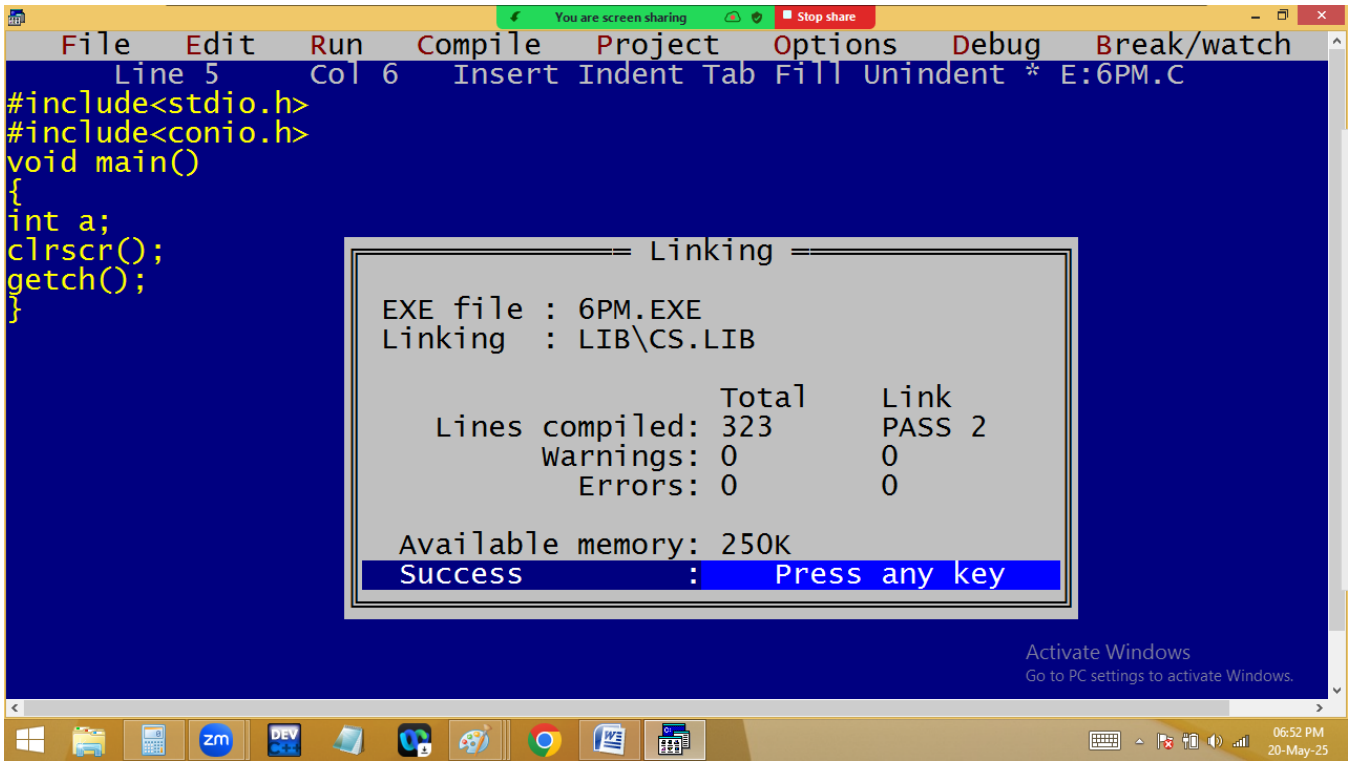


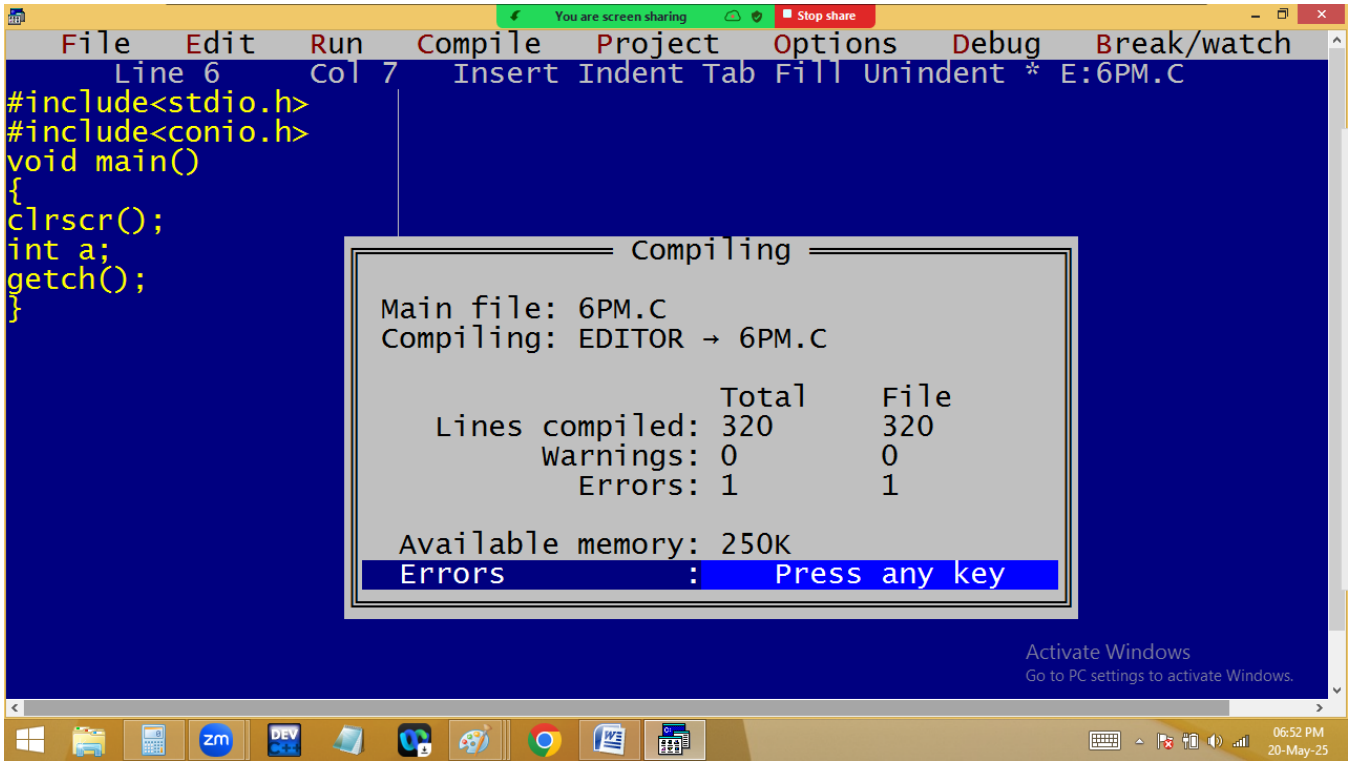

eg: 25 >> 5 = 0




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File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 20 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",25 & 15);
printf("%d\n",25 | 15);
printf("%d\n",25 ^ 15);
printf("%d\n",~25); /* n = -(n+1) */
printf("%d\n",~-25);
printf("%d\n",25<<1);
printf("%d\n",25<<2);
printf("%d\n",25<<15);
printf("%d\n",25<<16);
printf("%d\n",25>>1);
printf("%d\n",25>>2);
printf("%d\n",25>>5);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:51 PM
20-May-25


You are screen sharing
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9
31
22
-26
24
50
100
-32768
0
12
6
0
Activate Windows
Go to PC settings to activate Windows.
06:51 PM
20-May-25


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 6 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
getch();
}
Linking
EXE file : 6PM.EXE
Linking : LIB\CS.LIB
Total Link
Lines compiled: 323 PASS 2
Warnings: 0 0
Errors: 0 0
Available memory: 250K
Success : Press any key
Activate Windows
Go to PC settings to activate Windows.
06:52 PM
20-May-25


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 7 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
int a;
getch();
}
Compiling
Main file: 6PM.C
Compiling: EDITOR → 6PM.C
Total File
Lines compiled: 320 320
Warnings: 0 0
Errors: 1 1
Available memory: 250K
Errors : Press any key
Activate Windows
Go to PC settings to activate Windows.
06:52 PM
20-May-25

# Variables:

**It is a container is used to store the values in our programs.**

**Variable is a named memory where we can store and modify the values in our programs.**

**Always the variables are stored in primary memory i.e. RAM only. Due to this once the program execution is completed, automatically the variables are deleted from memory. Hence they are also called auto variable.**

**Variables are case sensitive. i.e. lower and upper are different.**

**Eg:**

**Int a=100;**

**Int A=200;**

**In C compiler the variables should be declared at the first line of any function. But in C++ / java etc we can declare anywhere.**

**Every variable having 2 stages.**

1. **Variable declaration / declared**
   **Eg: int a;**
2. **Variable initialization / definition**
   **Eg: a=100**

**Syntax:**

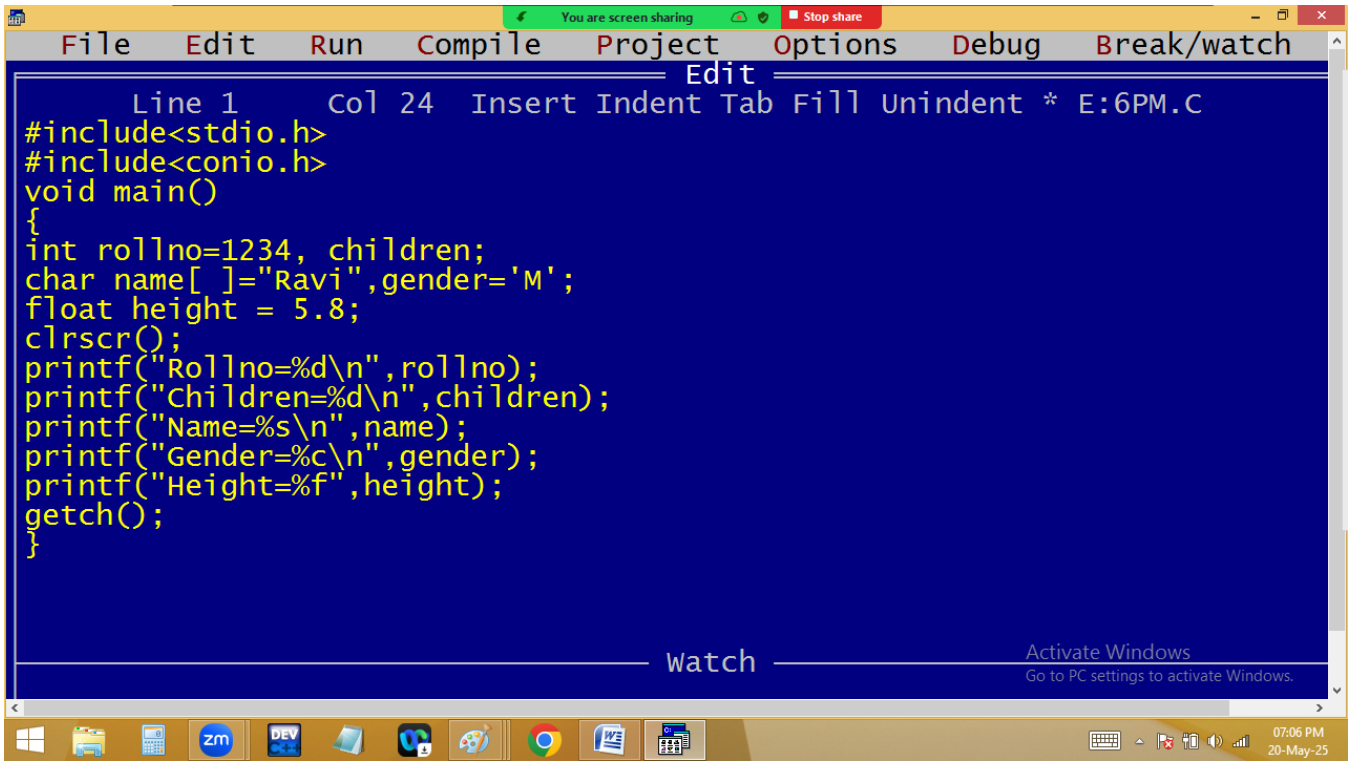
**Datatype variable[=value], var[=value],…;**

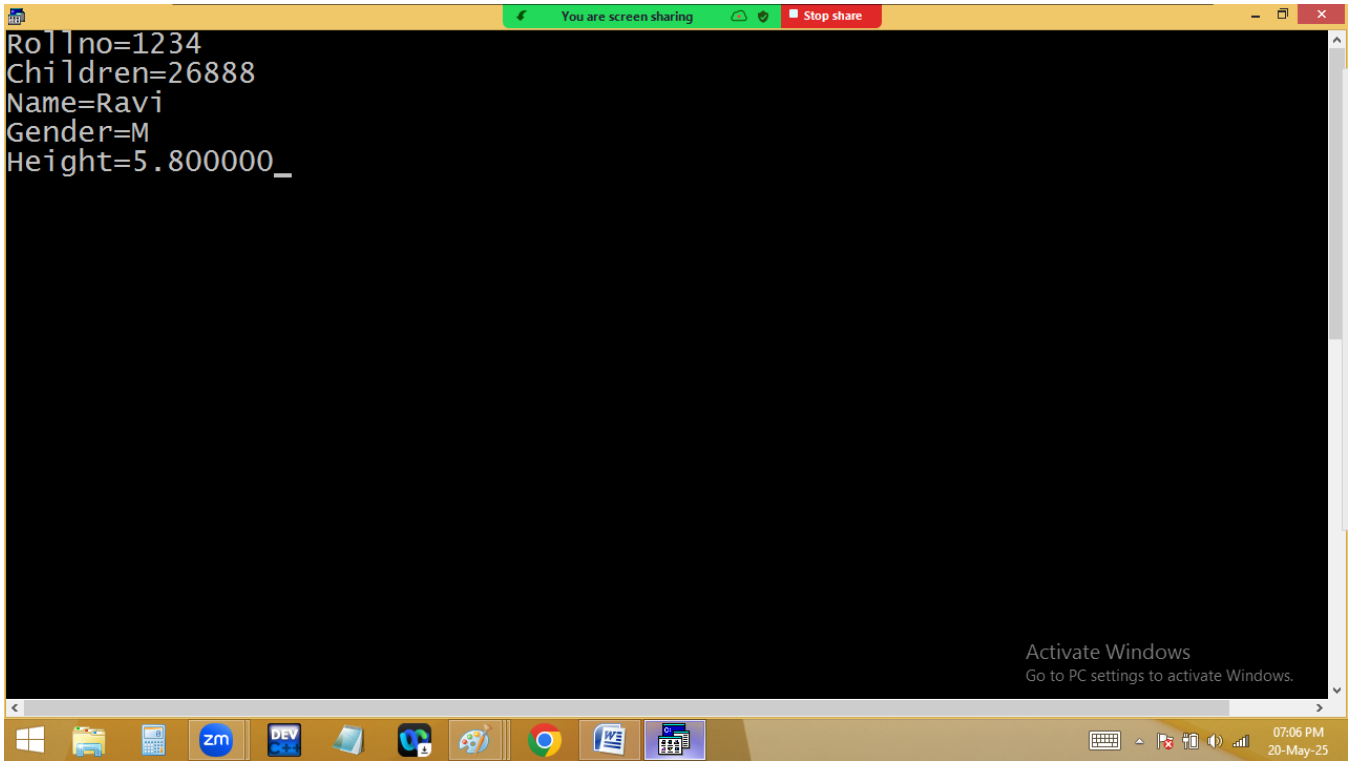
**Eg:**

**int rollno=1234, children;**

**char name[5]=”ravi”, gender=’M’;**

**float height = 5.8;**


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File Edit Run Compile Project Options Debug Break/watch
Edit
Line 1 Col 24 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int rollno=1234, children;
char name[ ]="Ravi",gender='M';
float height = 5.8;
clrscr();
printf("Rollno=%d\n",rollno);
printf("Children=%d\n",children);
printf("Name=%s\n",name);
printf("Gender=%c\n",gender);
printf("Height=%f",height);
getch();
}
Watch
Activate Windows
Go to PC settings to activate Windows.
07:06 PM
20-May-25


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Stop share
Rollno=1234
Children=26888
Name=Ravi
Gender=M
Height=5.800000
Activate Windows
Go to PC settings to activate Windows.
07:06 PM
20-May-25

# Memory allocation for variables:

4 BITS = 1 NIBBLE

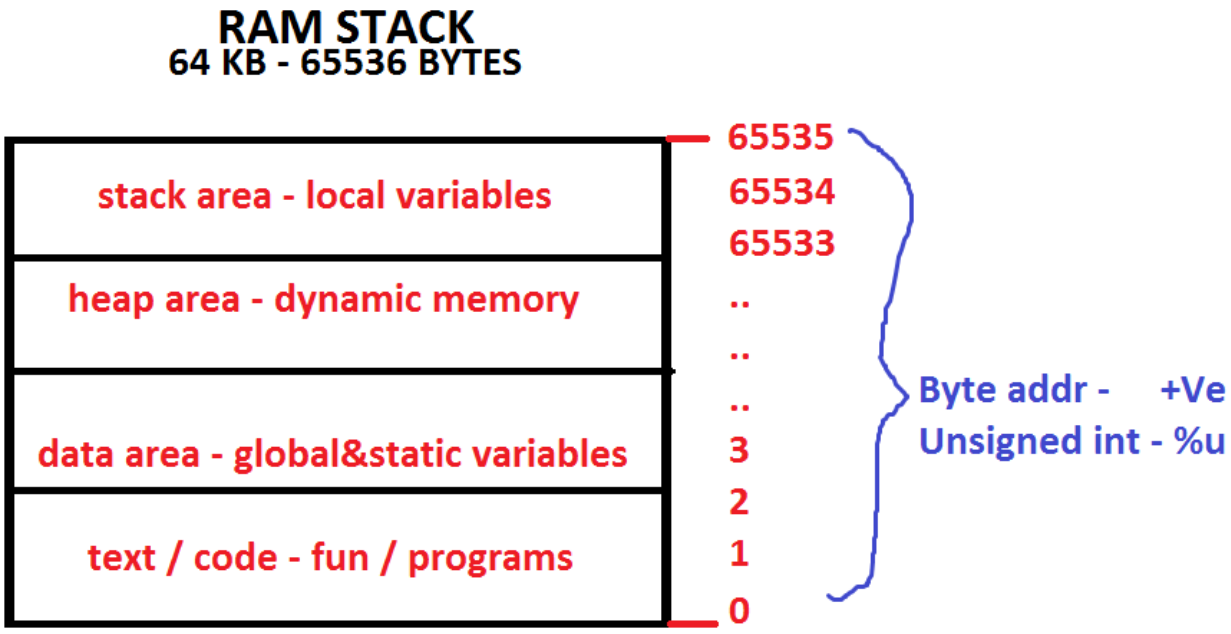
8 BITS = 1 BYTE

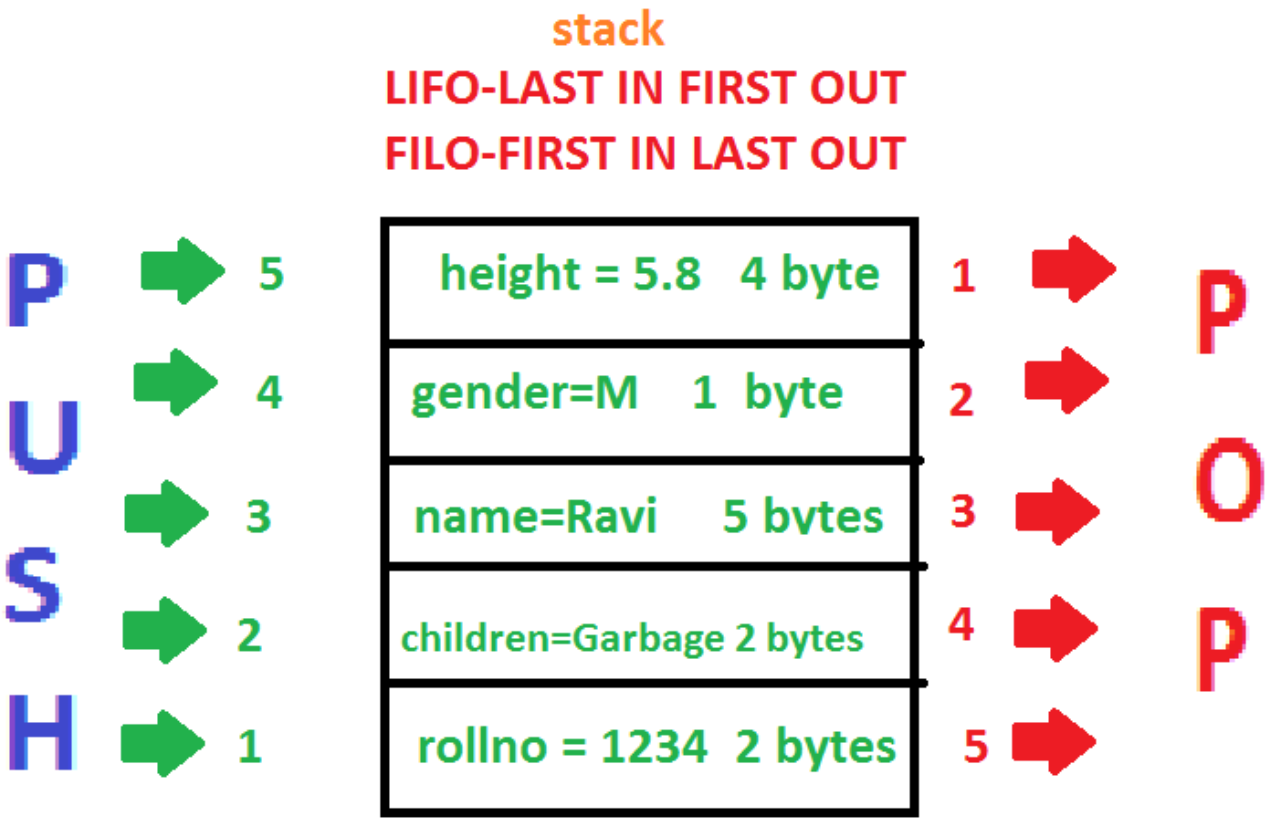
1024B = 1KB

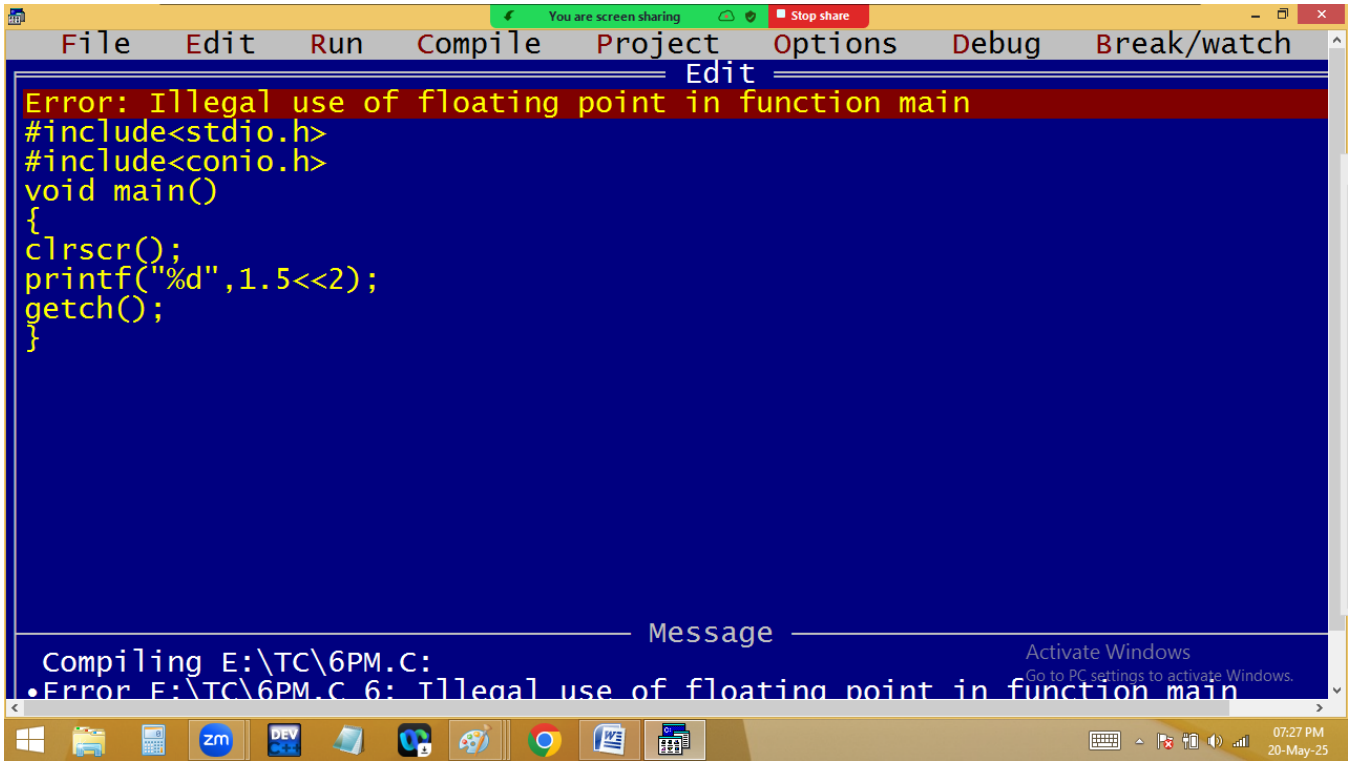
1024KB=1MB

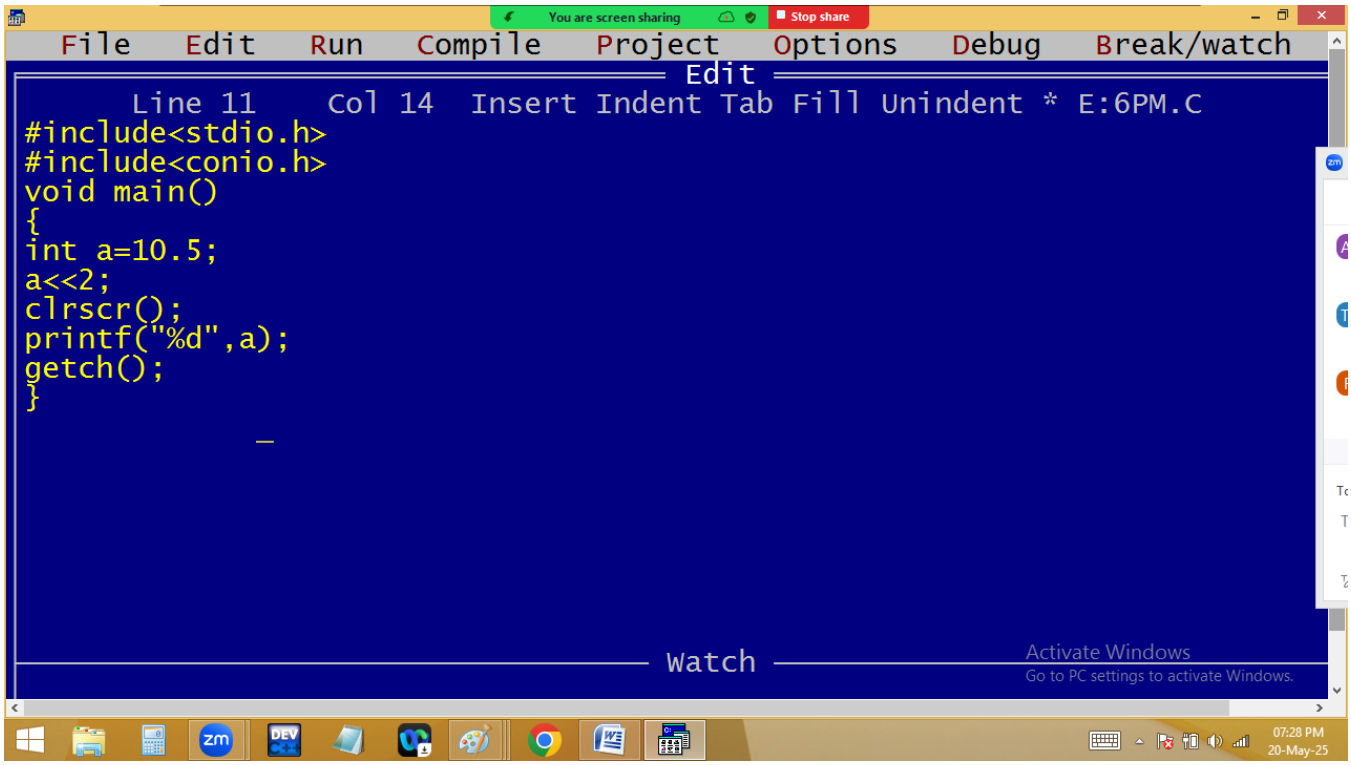
1024MB=1GB

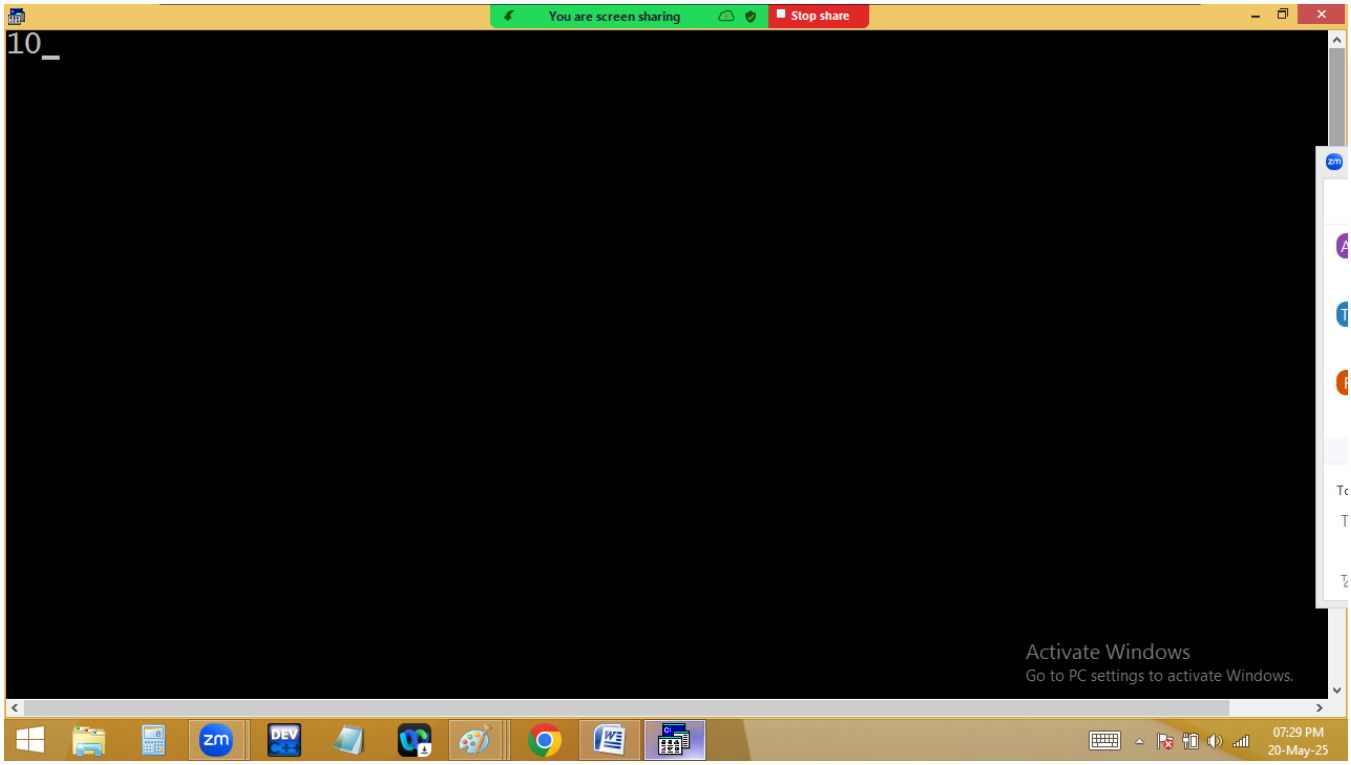
1024GB=1TB

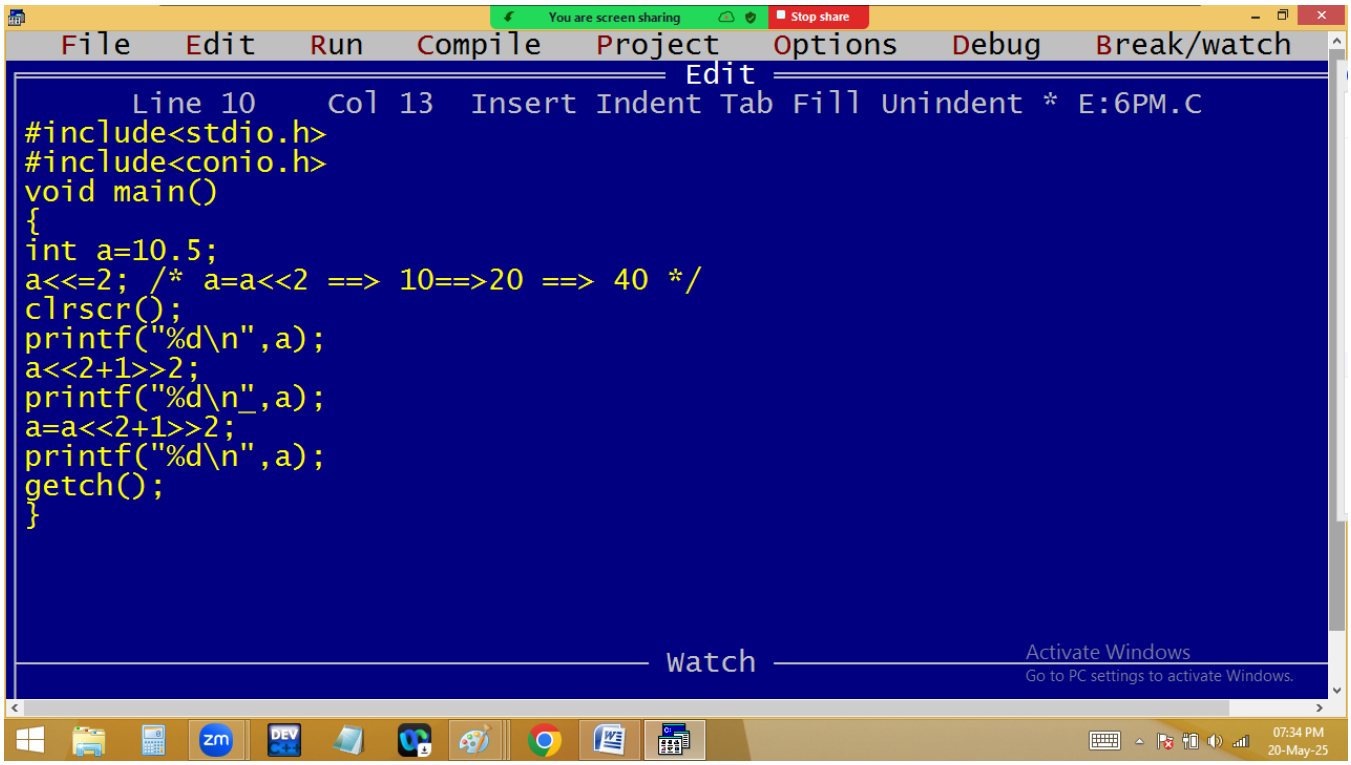



stack
LIFO-LAST IN FIRST OUT
FILO-FIRST IN LAST OUT
PUSH
5
4
3
2
1
height = 5.8 4 byte
gender=M 1 byte
name=Ravi 5 bytes
children=Garbage 2 bytes
rollno = 1234 2 bytes
1
2
3
4
5
POP


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File Edit Run Compile Project Options Debug Break/watch
Edit
Error: Illegal use of floating point in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d",1.5<<2);
getch();
}
Message
Compiling E:\TC\6PM.C:
Error E:\TC\6PM.C 6: Illegal use of floating point in function main
Activate Windows
Go to PC settings to activate Windows.
07:27 PM
20-May-25


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 14 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.5;
a<<2;
clrscr();
printf("%d",a);
getch();
}
Watch
Activate Windows
Go to PC settings to activate Windows.
07:28 PM
20-May-25

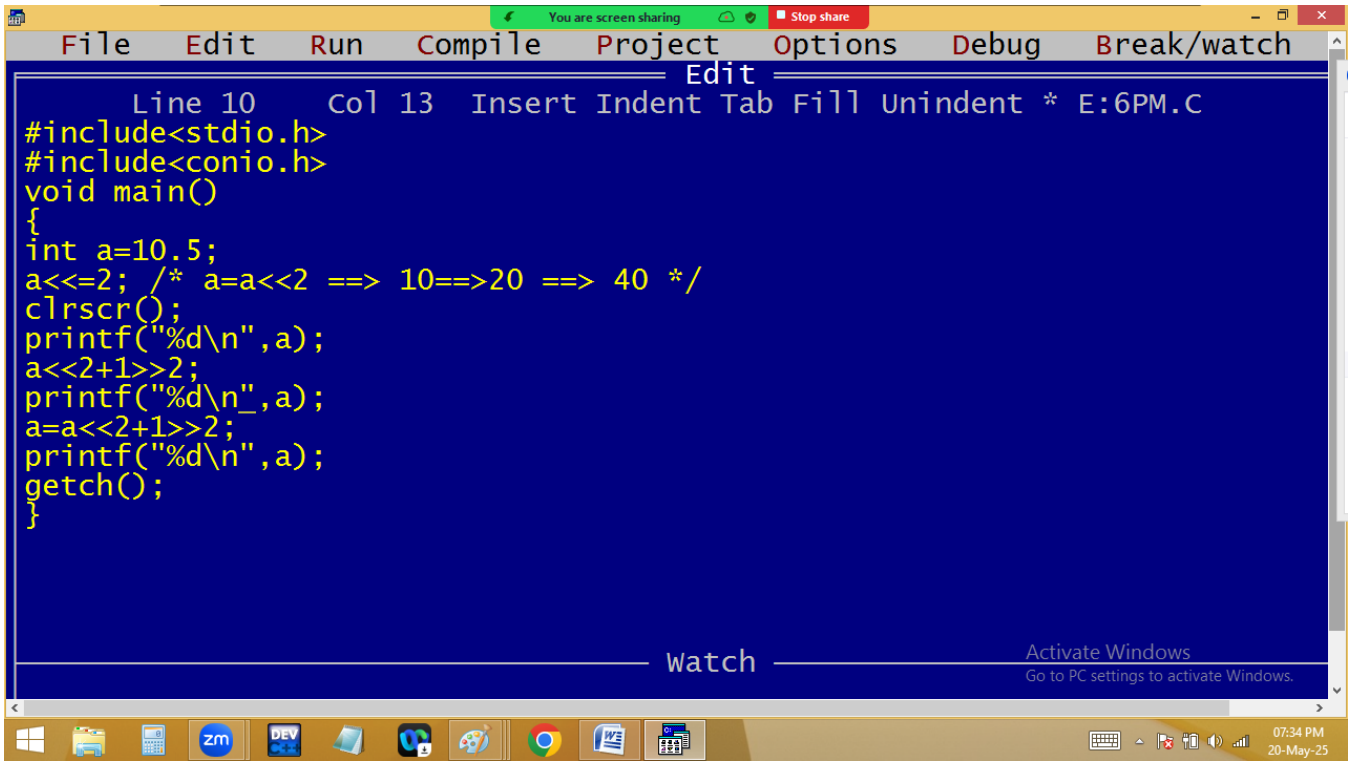

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Stop share
10
Activate Windows
Go to PC settings to activate Windows.
07:29 PM
20-May-25

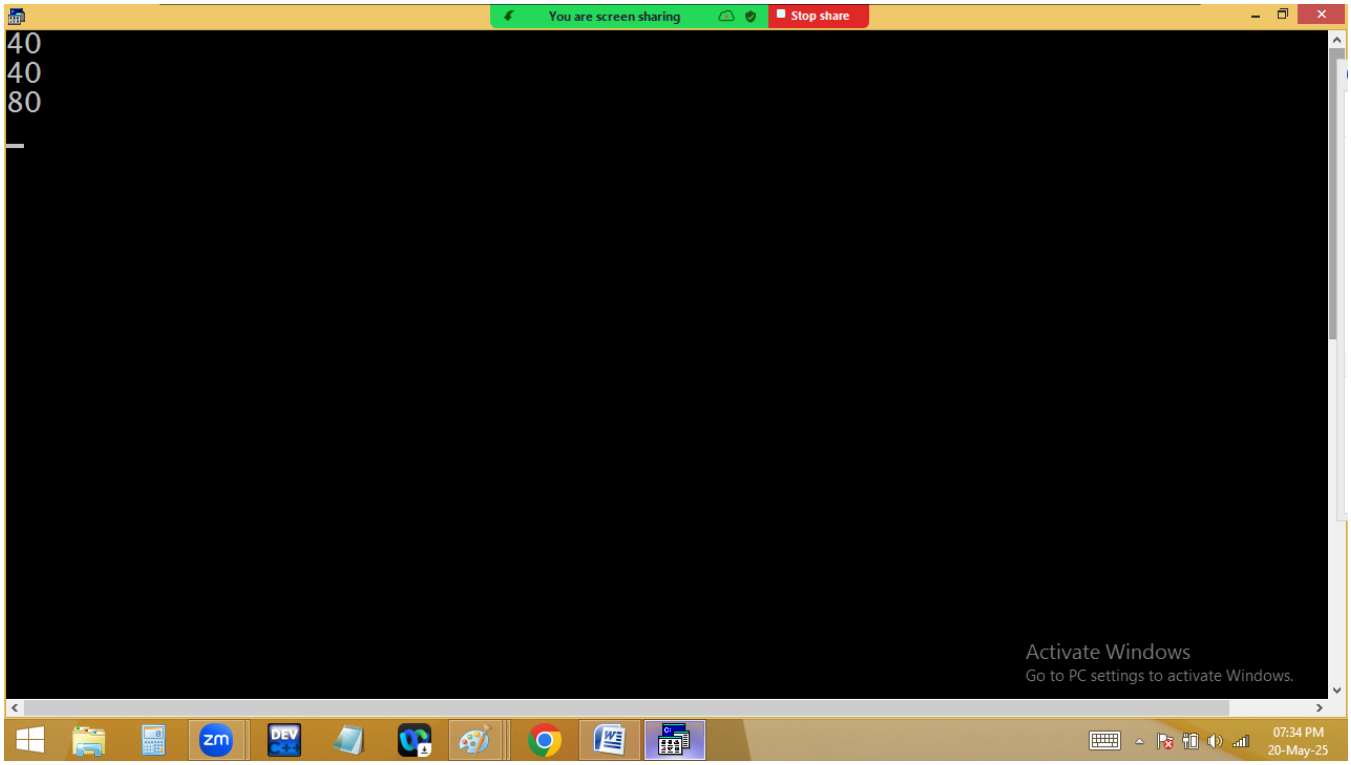



**int a = 10.5;**

**a<<2; ==> 10==> 20 ==> 40 [ = op not used ]**

**i.e. a=10**


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 10 Col 13 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.5;
a<<=2; /* a=a<<2 ==> 10==>20 ==> 40 */
clrscr();
printf("%d\n",a);
a<<2+1>>2;
printf("%d\n",a);
a=a<<2+1>>2;
printf("%d\n",a);
getch();
}
Watch
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
20-May-25


You are screen sharing
Stop share
40
40
80
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
20-May-25